Spatial-Contextual Information Utilization Framework for Land Cover Change Detection With Hyperspectral Remote Sensed Images

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***Abstract*— Land cover change detection (LCCD) using bitemporal remote sensing images is a crucial task for identifying the change areas on the Earth's surface. However, the utilization of hyperspectral remote sensing images (HRSIs) introduces challenges as the detection performance is affected by the spectral noise and deducing change detection accuracies. In this work, we concentrated on utilizing spatial-contextual information to improve the change detection performance while using HRSIs. First, a band selection approach is used to minimize the spectral redundancy of HRSIs. Second, an iterative spatial-adaptive filter is proposed to smooth the noise of HRSIs. Thereafter, the change magnitude between bitemporal HRSIs is measured by coupling change vector analysis (CVA) and the adaptive region around each pixel, resulting in a change magnitude image (CMI). Subsequently, the CMI is divided into a binary change detection map by using an Ostu threshold method. The experimental results on three pairs of real HRSIs efficiently demonstrated the feasibility and superiorities of the proposed approach compared with six state-of-the-art methods. For example, the improvement rates are approximately 0.43%–11.83% and 1.05%–15.41% for overall accuracy (OA) and average accuracy (AA), respectively. The code of our proposed approach will be available at: https://github.com/ImgSciGroup/2023-HSICD.**

***Index Terms*— Adaptive contextual analysis, hyperspectral remote sensing images (HRSIs), land cover change detection (LCCD).**

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I. Introduction

LAND cover change detection (LCCD) with remote sensing images plays an important role in acquiring the Earth's surface change, which is required for national land use map update [1], [2], landslide detection [3], object detection [4], [5], and environmental monitoring [6]. Various types of remote sensing images can be easily obtained for LCCD, including homogeneous optical images, hyperspectral images, and heterogeneous images, due to the development of remote sensing and aerial satellites. The utilization of a variety of images for LCCD has led to different challenges in practical applications [7], [8], [9]. The challenges and the corresponding solutions are reviewed in detail as follows.

A challenge for the bitemporal optical images used for LCCD is that these images are acquired at different dates, and the imaging process may be affected by sun heights, atmospheric conditions, and soil moisture, which will cause pseudo-changes and reduce the detection accuracies [10]. Various change detection methods have been promoted to address this challenge and improve the change detection performance. The most popular method is based on neighboring contextual information [11]. The fundamental concept of this method concentrated on the comparison and analysis of the relationship between a central pixel and its neighboring pixels, and the change magnitude is measured by the distance between the pairwise local areas [12], [13]. In brief, contextual information around a pixel can be described in two ways, namely, regular neighboring context [14], [15], [16] and adaptive neighboring context [17], [18], [19]. The mathematical model is another approach to explore contextual information for improving change detection performance, such as Markov random field [20], [21], [22], conditional random field [23], [24], [25], and level set theory [12], [26]. Various applications with real remote sensing images have verified the feasibility and superiorities of utilizing contextual information for improving change detection performance [27]. Although the contextual information-based approach concentrated on exploring the spatial–spectral features for detection, a universally applicable approach for all remote sensing images has not emerged due to the complex application scenario and the spectral uncertainty of the bitemporal remote sensing images. To date,

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the improvement space for LCCD with remote sensing images still remains [28].

Another way to address the challenges of change detection is to explore the deep features for measuring the change magnitude by using a neural network [29], [30]. Various neural networks have been widely used in LCCD with remote sensing images, such as convolutional neural network (CNN) [31], [32], [33], [34], recurrent neural network (RNN) [35], deep belief network [36], generative adversarial network [37], Hopfield neural network [38], and UNet++ [39]. To further improve the change detection performance with neural network, several backbones of neural networks have been coupled with attention mechanisms aimed at focusing on the most important parts of an image in a change detection task [40]. The network can explore the deep features, enhance the original feature's representation, and improve the learning efficiency of a network, such as a difference-enhancement dense-attention CNN [41], deeply supervised attention metric-based network [42], attention-guided end-to-end change detection network [43], attention mechanism-based deep supervision network [44], and spectral–spatial attention network based on change magnitude images (CMIs) [45], through attention mechanism. However, deep learning models require a large amount of annotated data for training. Moreover, labeling a substantial amount of training samples is time-consuming and labor intensive.

Comparing with change detection with optical remote sensing images, when hyperspectral remote sensing images (HRSIs) are used for LCCD, some additional challenges may be caused, such as high-dimensional characteristics that may result in a high computational complexity [46], and tiny spectral reflectance change captured by HRSIs may cause noise in a detection map [47]. The salt-and-pepper noise reduces the accuracies of change detection with HRSIs [9], [47]. Achieving LCCD with HRSIs has become important in applications with remote sensing images. These methods can be divided into binary change detection [48] and multiclass change detection [49]. When concentrating on the methodology of detection algorithms, these methods can be classified into traditional methods and deep learning-based approaches [9]. The characteristic of traditional LCCD methods lies in concentrating on how to measure the change magnitude between bitemporal HRSIs. Examples of these methods include change vector analysis (CVA) [50], compressed CVA (C2VA) [51], sequential-spectral CVA (S2CVA) [52], robust CVA (RCVA) [53], and normalized Euclidean distance (NED) [54]. These methods aimed at proposing a distance for measuring the change magnitude between bitemporal HRSIs. However, the different external factors (such as illumination and season difference) of bitemporal HRSIs and the redundancy bands of HRSIs typically cause detection noise in a change detection map. Several studies concentrated on feature transformation and smoothing noise in the detection map, such as multivariate alteration detection (MAD) [55], iteratively weighted MAD [56], principal component analysis (PCA) [57], temporal PCA (TPCA) [58], principal component analysis-means (PCA-Kmeans) [59], multiple morphological profiles [60], and slow feature analysis [61], to further improve the detection performance. In addition, collaborative sparsity nonlocal low-rank tensor [62], three-order tucker decomposition and reconstruction detector (TDRD) [63], and patch tensor-based change detection [64] are attractive for improving change detection accuracies while using HRSIs because some helpful features for LCCD with HRSIs may be enhanced in tensor space. Various studies with HRSIs indicated that feature exploration and transformation can smoothen noise and improve detection accuracies. However, feature exploration algorithms are typically complex for practical applications [9], [47], and no one method can be labeled as "good" or "bad" for different datasets. Therefore, the improvement space for LCCD with HRSIs is retained.

Besides traditional methods for LCCD with HRSIs, deep learning techniques have been successfully applied in the community of LCCD with HRSIs [65], [66], [67]. Du et al. [68] proposed an unsupervised deep slow feature extraction with deep learning for change detection with HRSIs. Wang et al. [48] and Yang et al. [69] built a general end-to-end 2D CNN framework and multiscale networks for hyperspectral images change detection, respectively. In addition, attention mechanism-based networks [66], [69], [70], [71], [72], [73], [74] have been widely used for LCCD with HRSIs. The attention mechanism has an advantage in focusing on the interesting parts of an image scene. The related applications indicated that deep learning networks can acquire satisfactory detection maps for change detection with HRSIs. However, the training network typically requires a large number of samples, and labeling samples for a pair of HRSIs is time-consuming and labor-intensive [65].

From the literature review above, some challenges of LCCD with HRSIs can be summarized as follows.

1) HRSIs contain a large number of bands, and the spectral redundancy may cause detection noise and deduce the detection accuracies. Although high-dimensional spectral bands can provide a detailed description of ground targets, it can also cause the occurrence of spectral mixing phenomenon and spectral redundancy. The physical characteristics of HRSIs usually have the effect that salt and pepper noise appears in a detection map based on such imagery [75], [76], [77]. Therefore, the challenge of smoothening noise and achieving a satisfied change detection map with HRSIs still persists in practical applications.
2) Spatial-contextual information has been verified as a factor that can improve the performance of change detection with remote sensed images [45], [78]. However, exploring and utilizing spatial-contextual information from images is challenging for LCCD with HRSIs. The changed and unchanged areas refer to various targets that have varying shapes and sizes. However, how to explore and utilize contextual information in an adaptive way for LCCD with HRSIs remains a challenge.
3) Promoting an effective distance for measuring the change magnitude between bitemporal HRSIs for LCCD

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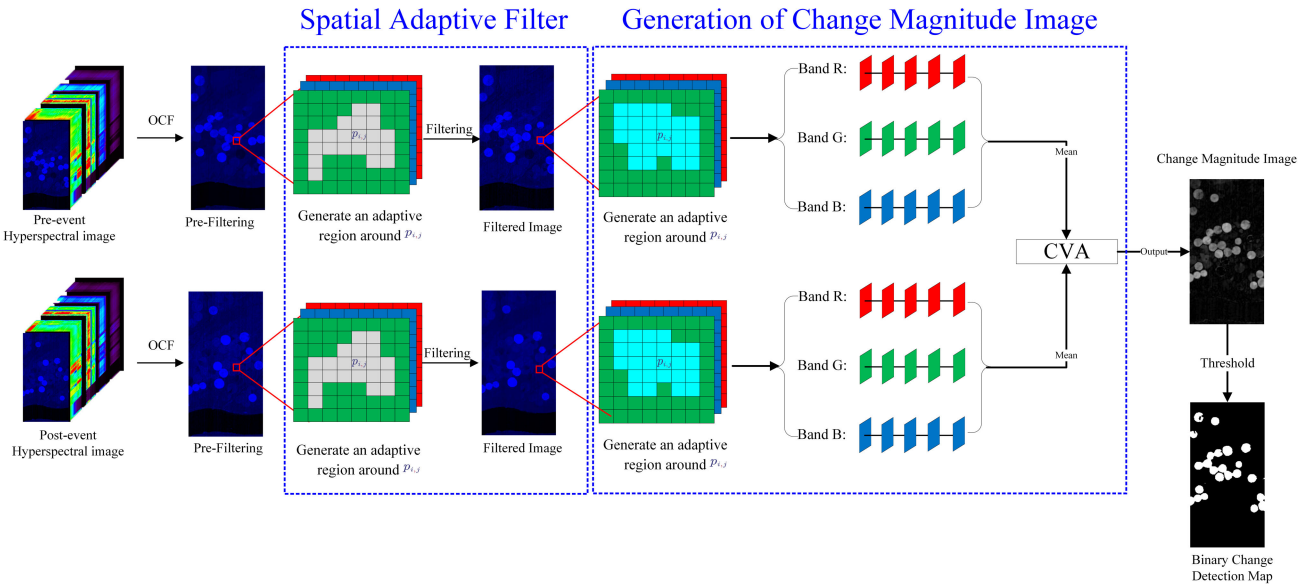


Fig. 1. Flowchart of the proposed hierarchical attention feature fusion-based network for LCCD with bitemporal RSIs.

is challenging. Similar to various LCCD with RSIs, bitemporal HRSIs are acquired at different external conditions, including sun height, soil moisture, and season phenology. The difference in external imaging conditions makes measuring the change magnitude complex [11], [27]. How to measure the change magnitude between bitemporal images is challenging for change detection with HRSIs.

In this work, we concentrate on proposing a spatial-contextual information utilization framework for improving the change detection performance with HRSIs. Our motivation lies in utilizing spatial-contextual information via an adaptive region around each pixel, including constructing an iterative spatial-adaptive filter and building a novel distance for measuring the change magnitude between pairwise images. The major contribution of the proposed approach can be briefly summarized as follows.

1) A novel iterative spatial-adaptive filter is proposed to smoothen the noise of HRSIs. In the proposed filter, in contrast with the spatial filter with a regular window or strict mathematical models, a spatial filter is constructed based on adaptive region. Moreover, the filtering progress is iterative to comprehensively improve the homogeneity in local areas.
2) A modified CVA (M-CVA) is proposed to measure the CMI. In the proposed M-CVA, two vectors are constructed based on the adaptive region around the corresponding central pairwise pixels to utilize spatial-contextual information.
3) Advanced detection accuracies are achieved by our proposed approach with three pairs of real HRSIs. In the experiments, the proposed approach achieved the best accuracies in terms of six widely used evaluation indicators compared with some traditional and deep learning methods. The comparisons efficiently demonstrated the advantages of the proposed approach.

The remainder of this article is organized as follows. The details of the proposed approach are presented in Section II. The experiences are provided in Section III, and the conclusions are drawn in Section IV.

II. Methodology

The proposed approach will be introduced in three parts, including an overview of the proposed approach, an iterative spatial-adaptive filter, and generation of a CMI with an M-CVA. The details are provided as follows.

A. Overview of the Proposed Approach

The flowchart of the proposed approach is shown in Fig. 1. First, a band selection algorithm named optimal clustering framework (OCF) [79] is adopted to select the top-three bands from the input HRSIs. Then, a spatial-adaptive filter is applied to smoothen the noise of the selected bands. Finally, an M-CVA based on the adaptive region is proposed to measure the change magnitude between the pairwise filtered images. When the entire bitemporal HRSIs are scanned and processed in this way, a CMI can be generated, and the Otsu [80] method is applied to the CMI to obtain the final binary change detection map.

B. Adaptive Region Extension of the Proposed Approach

Spectral noise is an inevitable phenomenon in HRSIs, and the noise of the bitemporal HRSIs typically causes a pseudo-change in a detection map. In this section, an iterative spatial-adaptive filter was proposed to smoothen noise and improve the local homogeneity of an image and the change detection performance with HRSIs. The details are presented as follows.

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First, an adaptive region around each pixel is generated for each band by measuring the spectral similarity between a pixel and its neighboring pixels. Here, a central pixel is denoted as $P_{i,j}$, and the standard deviation of the neighboring pixels around $P_{i,j}$ is denoted as $\mathrm{Std}_{i,j}$. Then, the spectral similarity between one pixel ($P_{i,j}$) and its eight-connected neighboring pixels ($P_{m,n}$) is measured by the condition $P_{m,n} \leq p_{i,j} \pm \mathrm{Std}_{i,j}$. If the condition is true, then $P_{m,n}$ will be labeled as the adaptive region's pixel; otherwise, it will not labeled as the adaptive region's pixels. After the eight-connected neighboring pixels are distinguished one by one, the labeled pixels will be selected and used to update the value $\mathrm{Std}_{i,j}$. Moreover, the labeled pixels will be used as central pixels for the next round of extension with the abovementioned rules. The extension around $P_{i,j}$ is terminated until no pixel can meet the defined condition. The eight-connected neighboring pixels around each pixel are initialized to calculate the standard deviation around the central pixel in the first iteration.

Second, the mean value of the pixels within an adaptive region around $P_{i,j}$ is used to replace the original value. All the pixels are processed and iteratively filtered, and the filtering progress is terminated when the spectral homogeneity of the adaptive regions from two adjacent iterations are approximate. The rule can be formalized with $\| \mathrm{Std}_{i,j}^{k} - \mathrm{Std}_{i,j}^{k+1} \| \leq \omega$, where $\mathrm{Std}_{i,j}^{k}$ and $\mathrm{Std}_{i,j}^{k+1}$ are the standard deviations of the adaptive regions around in the kth and $(k+1)$th iterations, respectively. $\| \mathrm{Std}_{i,j}^{k} - \mathrm{Std}_{i,j}^{k+1} \|$ is the Euclidean distance in terms of spectral feature.

Three aspects should be noted in the above iteration: 1) the adaptive region around each pixel was generated based on each selected band; 2) an adaptive spatial filter was applied on each pixel in an image scene; and 3) the iteration of the adaptive spatial filter will terminate until all the pixels in an image scene satisfied the condition ($\| \mathrm{Std}_{i,j}^{k} - \mathrm{Std}_{i,j}^{k+1} \| \leq \omega$).

C. Generation of CMI Based on the M-CVA

After the bitemporal HRSIs were filtered by the proposed iterative adaptive spatial filter, how to measure the change magnitude between the bitemporal filtered images remains a problem. To solve this problem, we assume that the two pixels in the bitemporal filtered images are $P_{i,j}$ and $P'_{i,j}$, and the two corresponding adaptive regions around the two pixels are denoted as $R_{ij}^{b_m}$ and $R_{ij}^{'b_m}$, where b_m denotes the spectral band and m is the band number. The distance between $R_{ij}^{b_m}$ and $R_{ij}^{'b_m}$ is used instead of the distance between $P_{i,j}$ and $P'_{i,j}$ calculated by the following equation:

$$\mathrm{CMI}_{i,j} = \sqrt{\sum_{m=1}^{m=3} \left(R_{i,j}^{b_m} - R_{i,j}^{'b_m} \right)^2} \tag{1}$$

where $\mathrm{CMI}_{i,j}$ is the change magnitude between the pairwise pixels at position (i, j) and $R_{i,j}^{b_m}$ and $R_{i,j}^{'b_m}$ are the adaptive regions around the pixel at position (i, j) for band b_m. When the entire image is scanned and processed with (1), the CMI between the bitemporal images was obtained.

Here, the generation of an adaptive region ($R_{i,j}^{b_m}$) around $P_{i,j}$ is clarified as follows. Given that the spectral distribution in filtered images is different from that of the corresponding original image, the adaptive region generation algorithm for the proposed filter used in Section II-B cannot be used directly to measure CMI. To address this problem, three rules are defined for constraining the extension of the adaptive region around a pixel: 1) the spectral similarity between a central pixel and its neighboring pixels should be less than T_1; 2) the number of the labeled pixel for an adaptive region around a central pixel should be less than T_2; and 3) the adaptive region around a pixel is iteratively extended until one of the above two rules is unsatisfactory.

D. Binary Change Detection Map

The aforementioned network is used as a basis to propose a compound loss function (CLF) for training the neural network. CLF contains four parts and each part has a different weight to balance the effect of different level features as shown in the following.

When obtaining a CMI, a binary threshold algorithm is required to acquire the binary change detection map. On the basis of the widely used Otsu's method in LCCD with RSIs [81], [82], Otsu's method [80] is adopted in the proposed approach to obtain a binary change detection map. As discussed in reference [80], the proposed Otsu's method, when applied to the binary change detection problem, yields a single threshold that separates pixels into two classes, namely, changed and unchanged classes. The threshold is determined by minimizing the intraclass intensity variance and maximizing the interclass variance.

III. Experiments

In this section, the experiments based on three pairs of HRSIs are used to evaluate the feasibility. The six state-of-the-art methods are compared to verify the superiority of the proposed approach. The details of the experiments are presented as follows.

A. Experiment Setting

Six widely used change detection algorithms, namely, CVA [50], support vector machine (SVM) [83], PCA-Kmeans [59], K-nearest neighbor (KNN) [84], general end-to-end 2-D CNN framework (GETNET) [48], and TDRD [63], are selected for comparison with the proposed method. The parameters of the comparative methods for each dataset are optimized by using the trial-and-error method. The details of the settings are shown in Table I. In addition, the description of each comparison method is presented as follows.

CVA [50]: This method generated a CMI by measuring the Euclidean distance between the bitemporal spectral vectors and then obtained the change result through threshold segmentation.

SVM [83]: This method saw the change detection task as a binary classification problem and adopted the SVM to obtain the binary change detection map.

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TABLE I
PARAMETER SETTING OF THE PROPOSED METHODS FOR EACH DATASET

Dataset	Method	Parameter
Farmland	SVM [83]	Sample=0.2
	PCA-Kmeans [59]	Block size=2
	KNN [84]	Sample=0.2
	GETNET [48]	Sample=0.7
	TDRD [63]	η=0.965
Hermiston	SVM [83]	Sample=0.2
	PCA-Kmeans [59]	Block size=2
	KNN [84]	Sample=0.2
	GETNET [48]	Sample=0.7
	TDRD [63]	η=0.99
Bay Area	SVM [83]	Sample=0.2
	PCA-Kmeans [59]	Block size=2
	KNN [84]	Sample=0.2
	GETNET [48]	Sample=0.7
	TDRD [63]	η=0.99

PCA-K-Means [59]: This method focused on obtaining CMI through PCA and the K-means method with $K = 2$. Then, the clustering average feature vector is adopted to calculate and obtain binary change detection map.

KNN [84]: This method aimed at exploring the nearest k samples to determine the label of the central pixel to obtain the change map.

GETNET [48]: This network obtained the abundance feature maps by applying linear and nonlinear spectral unmixing approach on each pixel and then put the abundance feature maps input to a CNN for processing to obtain the change results.

TDRD [63]: This method used a Tucker decomposer to decompose and reconstruct hyperspectral images, then used spectral angle analysis to measure the change magnitude between bitemporal hyperspectral images, and finally obtained the final result through a detector.

B. Data Description

Three pairs of HRSIs, namely, Farmland, Hermiston, and Bay Area, are used for the experiments, and the details of the used dataset are summarized in Table II. The description of the dataset is presented as follows.

The Farmland dataset consists of a pair of bitemporal HRSIs, which are acquired with Hyperion sensors installed on the EO-1 satellite on May 3, 2006 and April 23, 2007. In Fig. 2, the dataset refers to wetland agricultural land change events in Yancheng City, Jiangsu Province, China. The original dataset consists of 132 bands with a size of pixel and a 30-m/pixel spatial resolution.

The Hermiston dataset contained the bitemporal hyperspectral images, which are acquired with Hyperion sensors in 2004 and 2007, as shown in Fig. 3. This pair of images describes the crop change events in Elmiston City, OR, USA. The dataset has 224 bands with a size of pixels and a 30-m/pixel spatial resolution.

TABLE II
DETAIL DESCRIPTION FOR EACH DATASET

Dataset	Year	Location	Sensors	Size (pixel)	Resolution (m/pixel)
Farmland	2006 and 2007	Jiangu, China	Hyperion	220 × 430 × 132	30
Hermiston	2004 and 2007	Oregon, USA	Hyperion	390 × 200 × 224	30
BayArea	2013 and 2015	California, USA	AVIRIS	600 × 500 × 224	20

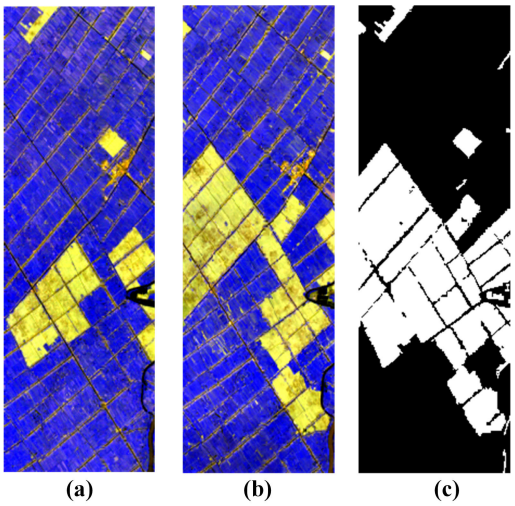


Fig. 2. Farmland dataset. (a) Image acquired on May 3, 2006. (b) Image acquired on April 23, 2007. (c) Ground truth.

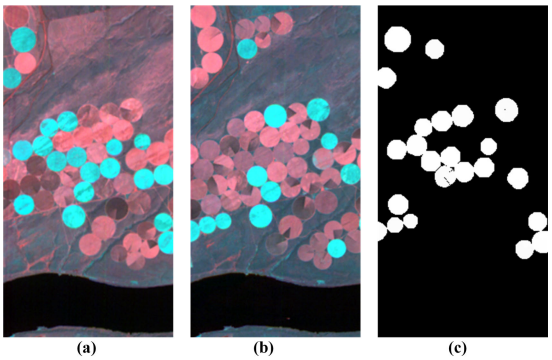


Fig. 3. Hermiston dataset. (a) Image acquired in 2004. (b) Image acquired in 2007. (c) Ground truth.

The Bay Area dataset was acquired in 2013 and 2015 with Airborne Visible InfraRed Imaging Spectrometer (AVIRIS) sensors. The research area is located in Patterson City, CA, USA (Fig. 4). The white areas in the ground truth map represent the changed areas, the gray areas denote the unchanged areas, and the black areas indicate the uninterested areas. This dataset consists of 224 bands, with a size of pixels, and the spatial resolution is 20 m/pixel.

C. Comparative Results

To achieve quantitative comparison, six widely used indicators were utilized to evaluate the detection results, as shown in Table III. Here, TP, TN, FP, and FN are represented as "true positive," "true negative," "false positive," and "false negative," respectively.

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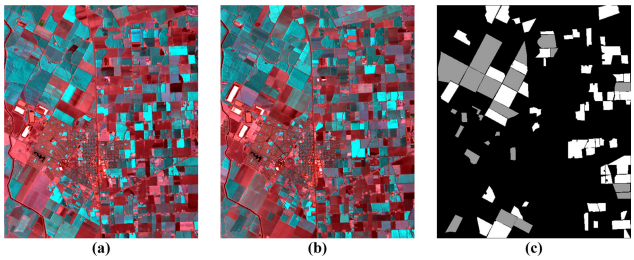


Fig. 4. Bay Area dataset. (a) Image acquired in 2013. (b) Image acquired in 2015. (c) Ground truth.

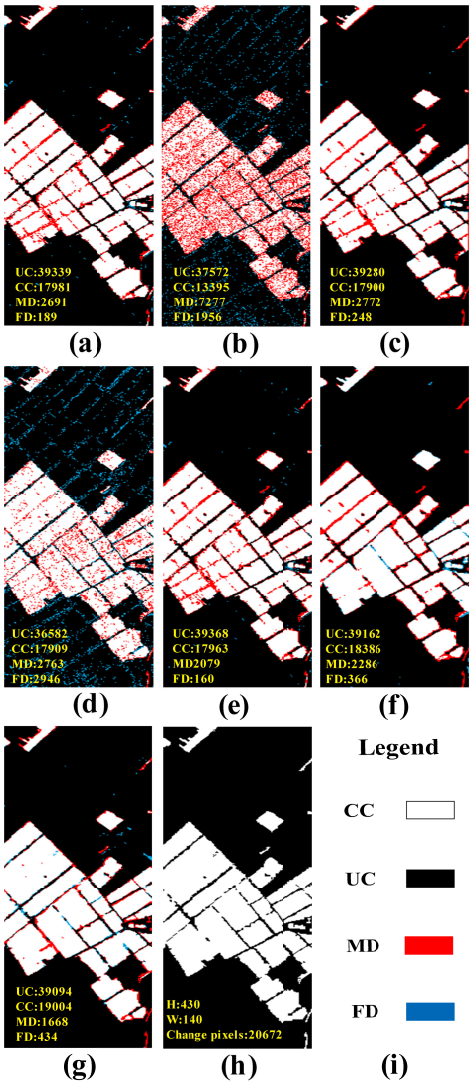


Fig. 5. Acquiring change maps using different methods for Data-Wetland: (a) CVA [50], (b) SVM [83], (c) PCA-Kmeans [59], (d) KNN [84], (e) GETNET [48], (f) TDRD [63], (g) proposed approach, (h) ground truth, and (i) legend.

1) Experimental Result on the Farmland Dataset: The proposed approach can obtain better performance compared with CVA [50], SVM [83], PCA-Kmeans [59], KNN [84], GETNET [48], and TDRD [63] in terms of accuracy in overall accuracy (OA), average accuracy (AA), Kappa, missing alarm (MA), and total error (TE), as shown by the bold font in Table III. The visual observation well supported the quantitative conclusion. Fig. 5 shows that the detection maps

TABLE III

QUANTITATIVE COMPARISON BETWEEN THE SELECTED METHODS AND THE PROPOSED APPROACH WITH THE OTHER METHODS FOR DATA-FARMLAND, KA $\in [-1, 1]$; THE OTHER VALUES ARE PRESENTED IN PERCENTAGE (%)

Method	OA	AA	Kappa	FA	MA	TE
CVA [50]	95.22	93.25	0.891	0.48	13.02	4.78
SVM [83]	84.66	79.92	0.638	4.95	35.20	15.34
PCA-Kmeans [59]	94.98	92.98	0.89	0.63	13.41	5.02
KNN [84]	90.52	89.59	0.790	7.45	13.37	9.48
GETNET [48]	95.32	93.25	0.891	**0.41**	13.1	4.77
TDRD [63]	95.59	94.01	0.900	0.93	11.06	4.41
Proposed method	**96.51**	**95.42**	**0.922**	1.10	**8.07**	**3.49**

TABLE IV

QUANTITATIVE COMPARISON BETWEEN THE SELECTED METHODS AND THE PROPOSED APPROACH WITH THE OTHER METHODS FOR DATA-HERMISTON, KA $\in [-1, 1]$; THE OTHER VALUES ARE PRESENTED IN PERCENTAGE (%)

Method	OA	AA	Kappa	FA	MA	TE
CVA [50]	92.41	**95.38**	0.862	1.57	7.68	1.91
SVM [83]	92.26	94.01	0.849	1.51	10.47	2.07
PCA-Kmeans [59]	91.30	95.85	0.799	2.99	5.85	3.02
KNN [84]	92.31	92.90	0.846	1.26	12.95	2.02
GETNET [48]	90.17	94.85	0.740	4.28	6.03	4.16
TDRD [63]	89.54	95.31	0.715	5.19	**3.60**	4.78
Proposed method	**92.64**	91.91	**0.865**	**0.66**	15.53	**1.68**

achieved by SVM and KNN have more salt-and-pepper noise than that of the detection map achieved by our proposed approach. In addition, the change detection map achieved by our proposed approach has fewer missed alarm pixels, as highlighted by red in Fig. 5(g).

2) Experimental Result on the Hermiston Dataset: The images described the change event about the crop growth trend difference caused by agricultural irrigation. Consequently, detecting a minor change between the bitemporal images is challenging. Based on the quantitative comparison in Table IV, the proposed approach achieved competitive accuracies in terms of OA, Ka, false alarm (FA), and TE. For example, the OA achieved by our proposed approach is 92.64%, which is the best accuracy, followed by OA = 92.41% that was achieved by CVA [50]. The Kappa achieved by the proposed approach is 0.865, which is similar to the Kappa for CVA (0.862). The visual comparison in Fig. 6 indicates that the detection map obtained by our proposed approach performs with less false alarm and missed alarm pixels. This phenomenon confirms the quantitative conclusion.

3) Experimental Result on the Bay Area Dataset: The proposed approach achieved advanced accuracies in terms of OA, Ka, and TE. In Table V, the proposed approach has OA = 97.52%, Ka = 0.881, and TE = 2.48%, which are the

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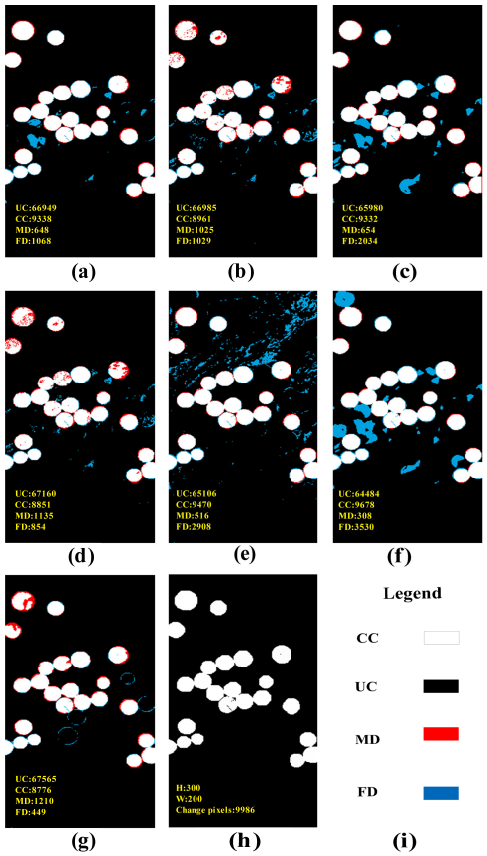


Fig. 6. Acquiring change maps using different methods for Data-Hermiston: (a) CVA [50], (b) SVM [83], (c) PCA-Kmeans [59], (d) KNN [84], (e) GETNET [48], (f) TDRD [63], (g) proposed approach, (h) ground truth, and (i) legend.

TABLE V

QUANTITATIVE COMPARISON BETWEEN THE SELECTED METHODS AND THE PROPOSED APPROACH WITH THE OTHER METHODS FOR DATA-BAY AREA, KA $\in [-1, 1]$; THE OTHER VALUES ARE PRESENTED IN PERCENTAGE (%)

Method	OA	AA	Kappa	FA	MA	TE
CVA [50]	96.18	86.77	0.809	0.66	25.8	3.82
SVM [83]	95.00	80.63	0.729	**0.17**	38.58	5.00
PCA-Kmeans [59]	96.55	**93.08**	0.846	2.29	**11.56**	3.45
KNN [84]	95.41	82.12	0.755	0.13	35.62	4.59
GETNET [48]	91.84	69.21	0.509	0.56	61.02	8.16
TDRD [63]	96.94	90.55	0.854	0.91	17.98	3.06
Proposed method	**97.52**	91.61	**0.881**	0.49	16.28	**2.48**

best performance when compared with those of CVA [50], SVM [83], PCA-Kmeans [59], KNN [84], GETNET [48], and TDRD [63]. In addition, the results achieved by SVM [83] and KNN [84] are subject to a large amount of noise, and many change pixels are missing in the detection map while using GETNET (Fig. 7) [48].

D. Discussion

The abovementioned experimental results indicated that the proposed method can achieve better performance in terms of

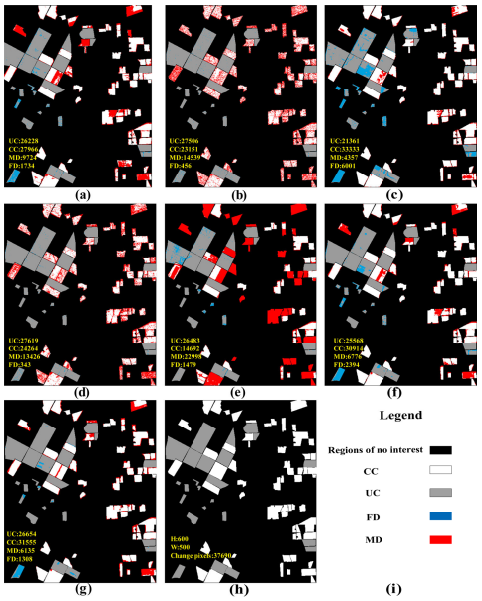


Fig. 7. Acquiring change maps using different methods for Data-Bay Area: (a) CVA [50], (b) SVM [83], (c) PCA-Kmeans [59], (d) KNN [84], (e) GETNET [48], (f) TDRD [63], (g) proposed approach, (h) ground truth, and (i) legend.

detection accuracy and visual observation compared with that using CVA [50], SVM [83], PCA-Kmeans [59], KNN [84], GETNET [48], and TDRD [63]. The relationship between the parameters (T_1 and T_2) of the proposed approach and the detection accuracies is investigated in this study to promote the extensive application of the proposed approach. The details are given as follows.

1) Sensitivity between T_1 and detection accuracies when T_2 is fixed: Fig. 8 shows that parameter T_1 exhibits distinct sensitivities when the proposed approach is applied to various datasets in terms of different evaluation indicators. For example, the accuracies in terms of OA, AA, and Ka for the Farmland dataset were decreased when T_1 increased from 5 to 100. By contrast, the accuracies were increased with the increment of $T_1 = 5$ to $T_1 = 100$ for the Hermiston dataset. When the accuracies of the Bay Area dataset were observed, the accuracies in terms of OA, AA, and Ka were decreased when T_1 was increased from 5 to 15. The accuracies exhibited a horizontal status with the increment of T_1 from 20 to 140. Furthermore, the relationship between T_1 and the accuracies in terms of FA, MA, and TE indicated that different datasets demonstrated varying sensitivities for the vibration of T_1 in terms of distinct indicators.
2) Sensitivity between T_2 and the detection accuracies when T_1 is fixed. Based on the investigation, T_1 was fixed at 10, 5, and 135 for the Farmland, Hermiston, and Bay Area datasets, respectively, when we observed the sensitivity between T_2 and detection accuracies. In Fig. 9, the OA of the Farmland and Hermiston datasets remained approximately 96.4% and 92.5%, respectively. However, the AA of the Bay Area dataset decreased from 90.25% to 91.61% when T_2 was increased from 5 to 140. The value of T_1 is large,

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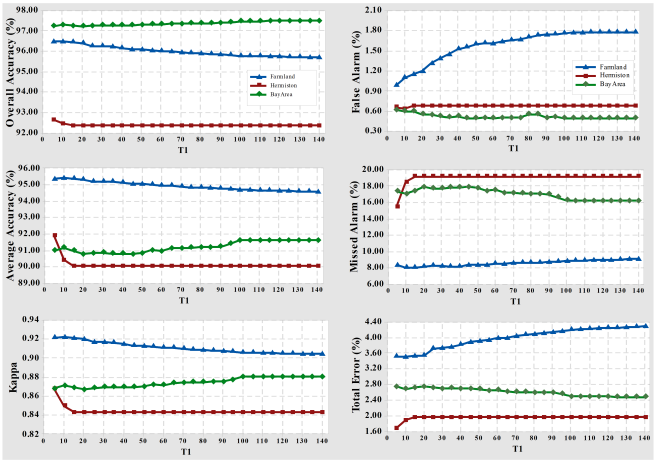


Fig. 8. Relationship between the detection accuracies and the variation of T_1 when T_2 was fixed at 20, 15, and 20 for the Farmland, Hermiston, and Bay Area datasets, respectively.

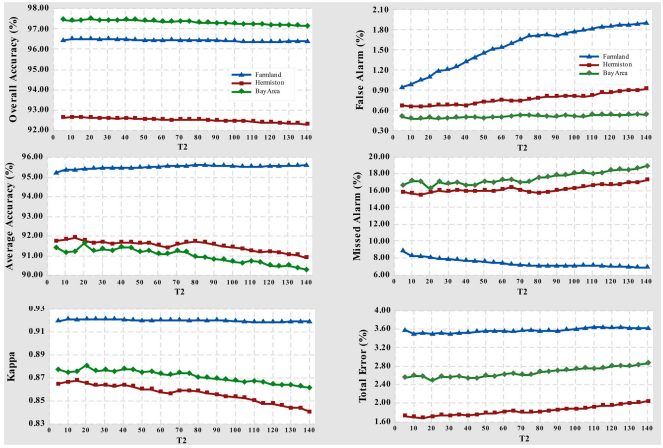


Fig. 9. Relationship between the detection accuracies and the variation of T_2 when T_1 was fixed at 10, 5, and 135 for the Farmland, Hermiston, and Bay Area datasets, respectively.

and more pixels were allowed to be assigned as the adaptive region around a pixel with the increment of T_2. Consequently, some pixels belonging to different targets may be wrongly assigned to one class. Moreover, the decreasing degrees varied for the different indicators and datasets compared with those of the AA and Ka in terms of the Hermiston and Bay Area datasets.

Based on the above discussion, distinct accuracy evaluation indicators show varying sensitivities for the diversification of T_1 and T_2 in terms of different datasets. Thus, the optimal combination of T_1 and T_2 for a given dataset and a specific application should obtain via trial-and-error experiments, which require a substantial amount of time.

IV. Conclusion

In this work, we concentrated on utilizing the spatial-contextual information to improve the change detection performance while using HRSIs. To achieve the objective, an iterative adaptive spatial filter is first proposed to smoothen the spectral noise of HRSIs, and an adaptive region was defined to explore contextual information around every pixel. In addition, a new distance was promoted through coupling with the adaptive region and CVA mechanism to measure the change magnitude between the pairwise bitemporal HRSIs. Finally, a simple yet widely used binary threshold named the Ostu approach was adopted to obtain a binary change detection map.

The experimental results based on three pairs of real HRSIs were evaluated for comparisons. In comparison with the six-related work, the quantitative evaluation and visual performance well indicated the feasibility and superiorities of the proposed approach.

In summary, although the proposed approach can obtain better performance and detection accuracies with three pairs of HRSIs, some limitations of the proposed approach remained. For instance, the setting of the parameters (T_1 and T_2) depended on the practitioner's experience in practical applications. Therefore, we will concentrate on promoting a parameter estimation method for predicting T_1 and T_2 for dynamically exploring spatial-contextual information in our future studies.

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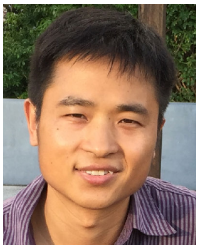
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